Experiment No. 2

Aim: Implementation of a given problem statement/s using classes and objects.

1. **Create a class called Employee that includes three pieces of information as instance variables: first name, last name, and monthly salary. Your class should have a constructor that initializes the three instance variables. Provide a setter and getter method for each instance variable. If the monthly salary is not positive, set it to 0.0. Write a test application named EmployeeTest that demonstrates the Employee class's capabilities. Create two Employee objects and display each object's yearly salary. Then give each Employee a 10% raise and display each Employee's yearly salary again.**

```
public class EmployeeTest {
    static class Employee {
        private String firstName;
        private String lastName;
        private double monthlySalary;

        public Employee(String firstName, String lastName, double monthlySalary) {
            this.firstName = firstName;
            this.lastName = lastName;
            if (monthlySalary > 0)
                this.monthlySalary = monthlySalary;
            else
                this.monthlySalary = 0.0;
        }
        public void setFirstName(String firstName) {
            this.firstName = firstName;
        }
```

```
    public String getFirstName() {
        return firstName;
    }
    public void setLastName(String lastName) {
        this.lastName = lastName;
    }
    public String getLastName() {
        return lastName;
    }
    public void setMonthlySalary(double monthlySalary) {
        if (monthlySalary > 0)
            this.monthlySalary = monthlySalary;
        else
            this.monthlySalary = 0.0;
    }
    public double getMonthlySalary() {
        return monthlySalary;
    }
    public double getYearlySalary() {
        return monthlySalary * 12;
    }
}

public static void main(String[] args) {
    Employee e1 = new Employee("Prasanna", "Bhurke", 20000);
    Employee e2 = new Employee("Rahul", "Sharma", 25000);

    System.out.println(e1.getFirstName() + " yearly salary: " + e1.getYearlySalary());
```

```
        System.out.println(e2.getFirstName() + " yearly salary: " + e2.getYearlySalary());

        e1.setMonthlySalary(e1.getMonthlySalary() * 1.10);
        e2.setMonthlySalary(e2.getMonthlySalary() * 1.10);

        System.out.println("After 10% raise:");

        System.out.println(e1.getFirstName() + " yearly salary: " + e1.getYearlySalary());
        System.out.println(e2.getFirstName() + " yearly salary: " + e2.getYearlySalary());
    }
}
```

```
PS D:\Prasanna\exp2> javac EmployeeTest.java
PS D:\Prasanna\exp2> java EmployeeTest
Prasanna yearly salary: 240000.0
Rahul yearly salary: 300000.0
After 10% raise:
Prasanna yearly salary: 264000.0
Rahul yearly salary: 330000.00000000006
PS D:\Prasanna\exp2>
```

2. **Implement a Java program to print the area of a rectangle by creating a class named 'Area' that has two methods. The first method, named 'setDim', takes the length and breadth of the rectangle as parameters. The second method, named 'getArea', returns the area of the rectangle. The length and breadth of the rectangle are entered through the keyboard**

```
import java.util.Scanner;
public class Area {
    private double length;
    private double breadth;
```

```
    public void setDim(double length, double breadth) {
        this.length = length;
        this.breadth = breadth;
    }
    public double getArea() {
        return length * breadth;
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Area rect = new Area();
        System.out.print("Enter length: ");
        double l = sc.nextDouble();
        System.out.print("Enter breadth: ");
        double b = sc.nextDouble();
        rect.setDim(l, b);
        System.out.println("Area of rectangle: " + rect.getArea());
        sc.close();
    }
}
```

```
PS D:\Prasanna\exp2> javac Area.java
PS D:\Prasanna\exp2> java Area
Enter length: 5
Enter breadth: 4
Area of rectangle: 20.0
PS D:\Prasanna\exp2>
```

3. **Write a Java program to demonstrate the use of static variables, static blocks, and static methods**

```
public class StaticDemo {
    static int count;
    static String message;
    static {
        count = 0;
        message = "Static block executed";
        System.out.println(message);
    }
    public StaticDemo() {
        count++;
    }
    static void showCount() {
        System.out.println("Objects created: " + count);
    }
    public static void main(String[] args) {
        StaticDemo s1 = new StaticDemo();
        StaticDemo s2 = new StaticDemo();
        StaticDemo.showCount();
```

```
    }
}
```

```
PS D:\Prasanna\exp2> javac StaticDemo.java
PS D:\Prasanna\exp2> java StaticDemo
Static block executed
Objects created: 2
PS D:\Prasanna\exp2>
```

4. **Write a Java program to implement a stack and a queue.**

```
import java.util.Scanner;
class Stack {
    int[] arr;
    int top;
    Stack(int size) {
        arr = new int[size];
        top = -1;
    }
    void push(int x) {
        if (top == arr.length - 1)
            System.out.println("Stack Overflow");
        else
            arr[++top] = x;
    }
    void pop() {
        if (top == -1)
            System.out.println("Stack Underflow");
        else
```

```
            System.out.println("Popped: " + arr[top--]);
    }
    void display() {
        if (top == -1)
            System.out.println("Stack Empty");
        else {
            for (int i = top; i >= 0; i--)
                System.out.print(arr[i] + " ");
            System.out.println();
        }
    }
}
class Queue {
    int[] arr;
    int front, rear;

    Queue(int size) {
        arr = new int[size];
        front = 0;
        rear = -1;
    }
    void enqueue(int x) {
        if (rear == arr.length - 1)
            System.out.println("Queue Overflow");
        else
            arr[++rear] = x;
    }
    void dequeue() {
```

```
        if (front > rear)
            System.out.println("Queue Underflow");
        else
            System.out.println("Deleted: " + arr[front++]);
    }
    void display() {
        if (front > rear)
            System.out.println("Queue Empty");
        else {
            for (int i = front; i <= rear; i++)
                System.out.print(arr[i] + " ");
            System.out.println();
        }
    }
}
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        Stack stack = new Stack(5);
        Queue queue = new Queue(5);

        while (true) {
            System.out.println("1.Push 2.Pop 3.Display Stack");
            System.out.println("4.Enqueue 5.Dequeue 6.Display Queue 7.Exit");
            int ch = sc.nextInt();

            switch (ch) {
```

```
            case 1:
                stack.push(sc.nextInt());
                break;
            case 2:
                stack.pop();
                break;
            case 3:
                stack.display();
                break;
            case 4:
                queue.enqueue(sc.nextInt());
                break;
            case 5:
                queue.dequeue();
                break;
            case 6:
                queue.display();
                break;
            case 7:
                sc.close();
                return;
            default:
                System.out.println("Invalid choice");
        }
    }
  }
}
```

```
PS D:\Prasanna\exp2> javac Main.java
PS D:\Prasanna\exp2> java Main
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
2
Stack Underflow
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
1
45
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
1
34
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
1
56
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
1
78
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
1
56
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
1
89
Stack Overflow
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
2
Popped: 56
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
5
Queue Underflow
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
4
56
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
4
67
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
4
87
1.Push 2.Pop 3.Display Stack
```

```
Stack Overflow
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
2
Popped: 56
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
5
Queue Underflow
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
4
56
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
4
67
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
4
87
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
4
12
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
4
34
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
4
11
Queue Overflow
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
3
78 56 34 45
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
6
56 67 87 12 34
1.Push 2.Pop 3.Display Stack
4.Enqueue 5.Dequeue 6.Display Queue 7.Exit
7
PS D:\Prasanna\exp2>
```

5. Write a Java program to arrange 10 names in alphabetical order.

```
import java.util.Scanner;
public class SortNames {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String[] names = new String[10];
        System.out.println("Enter 10 names:");
        for (int i = 0; i < 10; i++)
            names[i] = sc.nextLine();
        for (int i = 0; i < names.length - 1; i++)
            for (int j = 0; j < names.length - 1 - i; j++)
                if (names[j].compareToIgnoreCase(names[j + 1]) > 0) {
                    String temp = names[j];
                    names[j] = names[j + 1];
                    names[j + 1] = temp;
                }
        System.out.println("Names in alphabetical order:");
        for (String name : names)
            System.out.println(name);
        sc.close();
    }
}
```

```
PS D:\Prasanna\exp2> javac SortNames.java
PS D:\Prasanna\exp2> java SortNames
Enter 10 names:
b
a
y
r
fg
h
j
w
u
i
Names in alphabetical order:
a
b
fg
h
i
j
r
u
w
y
PS D:\Prasanna\exp2>
```